Ordinary Differential Equations (ODEs) & Linear Algebra

Welcome to the comprehensive course repository for **Ordinary Differential Equations (ODEs)**. This repository serves as a digitized, structured, and easily navigable study guide based on handwritten lecture notes. It covers the fundamental concepts, methods of solving various differential equations, and their applications.

Course Learning Outcomes (CLOs)

Based on the course syllabus, by the end of this course, students will be able to: * **CLO 1:** Describe the function of Differential Equations & systems of linear equations to explain physical situations. `[PLO 1]` * **CLO 2:** Apply appropriate methods to solve differential equations & systems of linear equations relevant to engineering. `[PLO 2]`

Recommended Textbooks

To supplement the materials in this repository, the following textbooks are recommended: 1. **Elementary Linear Algebra** by *Howard Anton* 2. **A First Course in Differential Equations** by *Dennis G. Zill* 3. **Advanced Engineering Mathematics** by *Erwin Kreyszig*

Sessional Marks Criteria

Assessment Type	Quantity	Total Marks	Notes
Midterm Exam	1	20	
Assignments	2	15	
Quizzes	2	5	Best 2 are counted

Repository Structure & Syllabus

This repository is organized into modular folders for easy navigation. Click on any module to explore the topics, concepts, and step-by-step solved mathematical problems.

Module 1: Fundamentals of Differential Equations

- Definitions of ODEs and PDEs
- Order and Degree of a Differential Equation
- Formation of Differential Equations by Eliminating Arbitrary Constants

Module 2: Solving First-Order ODEs

- Variable Separable Method
- Homogeneous Differential Equations
- Equations Reducible to Homogeneous Form
- Linear Differential Equations (Integrating Factors)
- Bernoulli's Equations (Reducible to Linear Form)

Module 3: Applications of First-Order ODEs

- Orthogonal Trajectories

Module 4: Higher-Order Linear ODEs (Constant Coefficients)

- Finding the Complementary Function (y_c)
- Particular Integral (y_p) - Exponential Functions
- Particular Integral (y_p) - Algebraic Functions
- Particular Integral (y_p) - Trigonometric Functions
- Particular Integral (y_p) - Product/Shift Rule

Module 5: Higher-Order Linear ODEs (Variable Coefficients)

- Cauchy-Euler Differential Equations

Note: The broader course syllabus also includes Linear Algebra, Partial Differential Equations (PDEs), and Fourier Series. This repository currently focuses primarily on the digitized lecture notes for 1st and 2nd/Higher Order ODEs.